

Levi Mackay

Computer Science Student, Open Source and Backend Systems

Rexburg, ID · levimackay@gmail.com · 208.284.4004 · levimackay.com · github.com/levimackay · linkedin.com/in/levi-mackay-217380396

Computer science student at BYU-Idaho, 3.98 GPA, graduating December 2027 and looking for a Summer 2027 software engineering internship. Maintains an open source AI coding agent with merged pull requests from developers I have never met, and ships production web work that local businesses pay for.

OPEN SOURCE

Lydia, a local AI coding agent

github.com/levimackay/lydia-cli · MIT · 36 stars, 8 forks

- A terminal coding agent that reads, edits, and tests code entirely against local Ollama models. No API keys, no subscription, and nothing leaves the machine. Roughly 6,500 lines of Python across 59 modules.
- 381 tests across 53 files run in GitHub Actions on every push, and the suite has to be green before anything merges.
- Two developers found the project on their own and opened pull requests that I reviewed and merged: Linux systemd user service support, and a fix for boolean config values stored as strings.

EXPERIENCE

Main Street Sites, Founder and Developer · BYU-Idaho Sandbox Incubator

Rexburg, ID · May 2026 to Present

- Building a self-serve web design studio for local businesses on Next.js 16 and TypeScript, with Neon Postgres behind Drizzle ORM, Clerk for auth, and Stripe for recurring billing.
- Shipped production redesigns for two Rexburg businesses, and ran the outreach and sales pipeline end to end rather than handing it off.

BYU-Idaho, Computer Science Teaching Assistant

Rexburg, ID · Apr 2026 to Present

- Work with students on programming fundamentals and debugging strategy, and grade assignments with written feedback rather than a score alone.
- Coach algorithmic thinking one on one, mostly by asking the next question instead of handing over the answer.

BYU-Idaho, CSE Student Employability and Outreach Assistant

Rexburg, ID · Apr 2026 to Present

Work with CS faculty on internship promotion, employer visits, and technical workshops, and track engagement numbers to find what actually brings students in.

SafeStreets, Dealer Operations and IT Specialist

Rexburg, ID · Mar 2025 to Aug 2025

Managed dealer account data, permissions, and system access, and resolved record discrepancies several teams depended on.

PROJECTS

RexNest, off campus housing marketplace

Next.js 16, Supabase Postgres, Stripe

- Ranks apartments on a weighted score over price, distance to campus, amenities, and resident reviews. Listings with fewer than three reviews fall back to the site average, so a place cannot ride on one or two ratings.
- 23 Postgres tables behind 129 lines of row level security policy. 123 tests across 10 files.
- Monetization built on Stripe: recurring subscriptions for property managers and one time paid boosts for listings. Billing tables are writable only by the Stripe webhook.

Serbo-Croatian dictionary and language reference

Original research · Python, Typst

- Wrote and compiled 5,709 dictionary entries from two years of my own handwritten field notes, collected while living in Croatia, Serbia, and Bosnia, alongside a 208 page scanned source.
- Built a Python and Typst pipeline to merge, index, and typeset the material into a 301 page reference. Anything the pipeline adds or corrects is flagged in the output instead of silently overwriting what I recorded.

EDUCATION

Brigham Young University Idaho

Sep 2024 to Dec 2027 (expected)

B.S. Computer Science · 3.98 GPA

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C# · **Web:** Next.js, React, Tailwind, three.js

Data: PostgreSQL, Supabase, Drizzle ORM, row level security, schema design · **Tooling:** Git, GitHub Actions, pytest, Stripe, Ollama